

Module-1

2 Marks Questions (Study Materials):

1. What is Machine Learning?
2. How Machine Learning is different from normal programming
3. Define Training and Test set.
4. What are the important objectives of machine learning?
5. What are the main activities involved in Machine Learning?
6. Mention the difference between Deep learning and Machine learning?
7. Mention the difference between Artificial Intelligence and Machine learning?
8. Mention the difference between Supervised and Unsupervised learning.
9. Which model learns and update itself through reward/ punishment?
10. Explain standardisation in data pre-processing?
11. What is Overfitting in Machine learning?
12. What is Underfitting in Machine learning?
13. Name the important steps involved to build a model in machine learning?
14. Explain How can you avoid underfitting?
15. What is the difference between quantitative and qualitative data?
16. Define Categorical data
17. Mention the essential sklearn library for data pre processing
18. Explain the terms Recall and Precision.
19. Mention the essential sklearn library for data pre processing
20. What is data splitting?
21. What is Cost function with respect to ML models?
22. What is Objective function with respect to ML models?
23. Define Reinforcement Learning?
24. What are Predictive models?
25. What are Descriptive Models?
26. How Predictive models differs from Descriptive Models?
27. What are sensitivity and specificity?
28. What do you mean by model parameter tuning in ML?

5 Marks Questions (Book Bank):

SHORT-ANSWER TYPE QUESTIONS (5 MARKS EACH):

- ✓ 1. What is human learning? Give any two examples. (5)
- ✓ 2. What are the types of human learning? Are there equivalent forms of machine learning?
- ✓ 3. What is machine learning? What are key tasks of machine learning? (5)
- ✓ 4. Explain the concept of penalty and reward in reinforcement learning. (5)
- ✓ 5. What are concepts of learning as search?
- ✓ 6. What are different objectives of machine learning? How are these related with human learning? (5)
- ✓ 7. Define machine learning and explain the different elements with a real example. (5)
- ✓ 8. Explain the process of abstraction with an example. (5)
- ✓ 9. What is generalization? What role does it play in the process of machine learning? (5)
- ✓ 10. What is classification? Explain the key differences between classification and regression? (5)
- ✓ 11. What is regression? Give example of some practical problems solved using regression? (5)
- ✓ 12. Explain the process of clustering in details. (5)
- ✓ 13. Write short notes on any two of the following: (5)
 - (a) Application of machine learning algorithms
 - (b) Supervised learning
 - (c) Unsupervised learning
 - (d) Reinforcement learning

- ✓ 1. What are the main activities involved when you are preparing to start with modelling in machine learning? (5)
- ✓ 2. What are the basic data types in machine learning? Give an example of each one of them. (5)
- ✓ 3. Differentiate: (5)
 - (a) Categorical vs. Numeric attribute
 - (b) Dimensionality reduction vs. Feature selection
- ✓ 4. Write short notes on any two: (5)
 - ✗ (a) Histogram
 - ✗ (b) Scatter plot
 - ✓ (c) PCA
- ✓ 5. Why do we need to explore data? Is there a difference in the way of exploring qualitative data vis-a-vis quantitative data? (5)
- ✗ 6. ~~What are different shapes of histogram? What are bins?~~
- ✓ 7. How can we take care of outliers in data? (5)
- ✓ 8. What are the different measures of central tendency? Why do mean, in certain data sets, differ widely from median? (5)
- ✗ 9. ~~Explain how bivariate relationships can be explored using scatter plot. Can outliers be detected using scatter plot?~~
- ✗ 10. ~~Explain how cross-tabs can be used to understand relationship between two variables~~

- ✓ 1. What is a model in context of machine learning? How can you train a model? (5)
- ✗ 2. ~~Explain "No-Free-Lunch" theorem in context of machine learning.~~
- ✓ 3. Explain, in details, the process of K -fold cross-validation. (5)
- ✓ 4. Explain the bootstrap sampling. Why is it needed? (5)
- ✗ 5. ~~Why do we need to calculate Kappa-value for a classification model? Show, with a sample set of data, how to calculate Kappa-value of a classification model.~~
- ✓ 6. Explain the process of ensemble of models. What role does it play in machine learning? (5)
- ✓ 7. What is the main purpose of a descriptive model? State some real-world problems solved using descriptive models. (5)
- ✓ 8. Explain the process of evaluating a linear regression model. (5)
- ✓ 9. Differentiate (any two): (5)
 - Predictive vs. descriptive models
 - Model underfitting vs. overfitting
 - Cross-validation vs. bootstrapping
- ✓ 10. Write short notes on any two: (5)
 - (a) LOOCV
 - (b) F -measure
 - (c) Silhouette width
 - (d) ROC curve

5 Marks Questions (Study Materials):

1. What are the basic design issues and approaches to machine learning?
2. Differentiate between Supervised, Unsupervised and Reinforcement Learning.
3. Explain the issues involved in Machine Learning models.
4. Explain the ROC curve with a diagram.
5. Discuss the steps involved in data preparation.
6. Discuss the difference between Standardization and Normalization in data processing.
7. Explain the Overfitting and Underfitting issues.
8. Discuss the trade-off between Bias and Variance Machine Learning
9. Explain synthetic data and where it is used?
10. Why overfitting happens?
11. Why underfitting happens?
12. Which is more important to you: model accuracy or model performance?
13. What evaluation approaches would you work to gauge the effectiveness of a machine learning model?
14. Which is more important to you: model accuracy or model performance?

10 Marks Questions (Book Bank):

LONG-ANSWER TYPE QUESTIONS (10 MARKS QUESTIONS):

- ✓ 1. What is machine learning? Explain any two business applications of machine learning. What are the possible ethical issues of machine learning applications? (10)
 - ✓ 2. Explain how human learning happens: (10)
 - (a) Under direct guidance of experts
 - (b) Under indirect guidance of experts
 - (c) Self-learning
 - ✓ 3. Explain the different forms of machine learning with a few examples. (10)
 - ✓ 4. Compare the different types of machine learning. (10)
 - ✓ 5. What do you mean by a well-posed learning problem? Explain important features that are required to well-define a learning problem. (10)
 - ✓ 6. Can all problems be solved using machine learning? Explain your response in detail. (10)
 - ✓ 7. What are different tools and technologies available for solving problems in machine learning? Give details about any two of them. (10)
 - ✓ 8. What are the different types of supervised learning? Explain them with a sample application in each area. (10)
 - ✓ 9. What are the different types of unsupervised learning? Explain their difference with a sample application in each area. (10)
 - ✓ 10. Explain, in details, the process of machine learning. (10)
 - (a) Write short notes on any two: (10)
 - ✓ ~~(i) MATLAB~~
 - (ii) Application of machine learning in Healthcare
 - (iii) Market basket analysis
 - (iv) Simple linear regression
 - ✓ (b) Write the difference between (any two): (10)
 - (i) Abstraction and generalization
 - (ii) Supervised and unsupervised learning
 - (iii) Classification and regression
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- ✓ 1. What are the main activities involved in machine learning? What is meant by data pre-processing? (10)
 - ✓ 2. Explain qualitative and quantitative data in details. Differentiate between the two. (10)
 - ✓ 3. Prepare a simple data set along with some sample records in it. Have at least one attribute of the different data types used in machine learning. (10)
 - ✓ 4. What are the different causes of data issues in machine learning? What are the fallouts? (10)
 - ✓ 5. Explain, with proper example, different ways of exploring categorical data.
 - ✓ 6. When there are variables with certain values missing, will that impact the learning activity? If so, how can that be addressed?
 - ✓ 7. Explain, in details, the different strategies of addressing missing data values.
 - ✓ 8. What are the different techniques for data pre-processing? Explain, in brief, dimensionality reduction and feature selection. (10)

9. (a) What is IQR? How is it measured?
 (b) Explain, in details, the different components of a box plot? When will the lower whisker be longer than the upper whisker? How can outliers be detected using box plot? (10)
10. (a) Write short notes on any two:
 (i) Interval data
 (ii) Inter-quartile range
 (iii) Cross-tab (10)
- (b) Write the difference between (any two): (10)
 (i) Nominal and ordinal data
 (ii) Box plot and histogram
 (iii) Mean and median
1. What is a target function? Express target function in context of a real-life example. How is the fitness of a target function measured? (10)
2. What are predictive models? What are descriptive models? Give examples of both types of models. Explain the difference between these types of models. (10)
3. Explain, in details, the process of evaluating the performance of a classification model. Explain the different parameters of measurement. (10)
4. (a) What is underfitting in context of machine learning models? What is the major cause of underfitting?
 (b) What is overfitting? When does it happen?
 (c) Explain bias-variance trade-off in context of model fitting. (10)
5. Can the performance of a learning model be improved? If yes, explain how. (10)
6. How would you evaluate the success of an unsupervised learning model? What are the most popular measures of performance for an unsupervised learning model? (10)
7. Is there a way to use a classification model for a numerical data or a regression model on a categorical data? Explain your answer. (10)
8. Describe the process of predictive modelling for numerical values. How is it different from predictive modelling for categorical values? (10)
9. While predicting malignancy of tumour of a set of patients using a classification model, following are the data recorded:
 (a) Correct predictions – 15 malignant, 75 benign
 (b) Incorrect predictions – 3 malignant, 7 benign
 Calculate the error rate, Kappa value, sensitivity, precision, and F-measure of the model. (10)
10. (a) Write short notes on any two:
 (i) Holdout method
 (ii) 10-fold cross-validation
 (iii) Parameter tuning (10)
- (b) Write the difference between (any two): (10)
 (i) Purity vs. Silhouette width
 (ii) Bagging vs. Boosting
 (iii) Lazy vs. Eager learner

10 Marks Questions (Study Materials):

1. Explain the stages involved in building a model in machine learning.
2. Discuss the performance matrices in model evaluation.
3. Describe the applications of machine learning.
4. Discuss types of data in machine learning
5. Explain the dimensionality reduction in data preprocessing.